PAThrive: A Web-Based Training Management and Skills Utilization Tracking System for the Office of the Extension Services of Southern Luzon State University

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CHAPTER I

INTRODUCTION

In recent years, rapid developments in information and communication technologies (ICT) have reshaped the technological landscape, particularly through digital transformation trends that emphasize automation of administrative processes, cloud-based platforms, and data analytics integration across higher education and community organizations (World Bank, 2023). Institutions worldwide are adopting these innovations to enhance operational efficiency, improve accessibility for diverse stakeholders, and ensure greater transparency in service delivery, enabling real-time decision-making and scalable program management essential for modern workforce development.

Recent studies underscore the effectiveness of similar electronic document and training management systems. For instance, Daluyon (2024) demonstrated how web-based platforms reduced administrative processing time by 65% in Philippine universities through automated workflows, though limitations persisted in user adoption and mobile compatibility. In a similar manner, Castro et al. (2022) found that ICT-driven tracking systems improved reporting accuracy in extension services by 78%, yet highlighted gaps in post-implementation support and integration with legacy data, revealing opportunities for more comprehensive, user-centric solutions tailored to institutional needs.

Southern Luzon State University (SLSU), a public State University and College (SUC) in the Calabarzon region, operates its Office of Extension Services to coordinate university-wide extension programs across various departments and community sectors. These programs aim to equip community members with livelihood skills and technical competencies in order to help the beneficiaries prepare for their potential future livelihood jobs while fostering partnerships with local stakeholders. According to the conducted interview by the researchers, faculty members from different departments serve as trainers while extension coordinators encourage trainers to provide training proposal, manage post-training monitoring and documentation. Trainings often target specific beneficiary groups, such as out-of-school youth, displaced workers, solo parents or individuals referred by partner local government units, ensuring participants meet the qualifications for the training offered. Trainers are responsible for delivering instruction and recording training materials, while extension coordinators compile post-training evaluations and impact assessment summaries to ensure accurate reporting.

Based on the conducted interview, the current process of managing training implementation relies heavily on paper-based documents, messenger application, and email for communication, as well as separately stored digital files. Trainers usually use hard-copy material for modules, and post-training evaluations, while coordinators consolidate impact assessment results from multiple sources. This fragmented approach can be time-consuming, prone to errors, and often results in delays when preparing accomplishment reports or monitoring post-training outcomes. The manual nature of data collection and storage limits the ability of administrators to generate timely and accurate summaries or track participants' skill utilization effectively. These challenges highlight the need for a more centralized and structured system that ensures organized and accessible documentation of training implementation and post-training activities. The main issue lies in how the information from these trainings is handled. Currently, there is no single online platform where staff can enter training details, upload instructional materials, record evaluations, and submit impact assessments in one place. Instead, records are kept in separate files or on paper, which slows down processes, makes it harder to access information, and increases the chance of mistakes.

In response to these challenges, this study proposes the development of PAThrive, a web-based Training Management and Skills Utilization Tracking System. The system enables authorized users to record approved training details, upload modules and instructional materials, encode post-training evaluations, and submit impact assessment summaries. It also includes an announcement feature for posting updates on ongoing or completed trainings, as well as a limited public viewing function. Through this public viewing feature, external stakeholders can access selected completed trainings and accomplishments, and inquiries can be sent directly to the Extension Office via the Contact Us page. These features help promote transparency, increase institutional visibility, and encourage potential partnerships for future training programs while still maintaining the security of internal data. Focusing exclusively on training implementation and post-training monitoring helps PAThrive organize records efficiently while leaving training planning and approvals outside its scope.

Providing a centralized digital platform like PAThrive contributes to the operational effectiveness of the Office of Extension Services. Structured data allows administrators to monitor training activities, evaluate program outcomes, and generate accurate reports, improving both efficiency and accountability. The system also aligns with larger digital transformation initiatives in higher education, fostering innovation in managing extension programs. From a sustainability perspective, reducing paper-based processes supports responsible resource management, while public access to select information promotes transparency and strengthens partnerships with local governments and community organizations. Furthermore, PAThrive contributes to the United Nations Sustainable Development Goals (UNSDGs) by supporting SDG 4: Quality Education, through effective skills development and training programs; SDG 9: Industry, Innovation, and Infrastructure, by demonstrating innovation in ICT adoption; and SDG

17: Partnerships for the Goals, by encouraging collaboration with external stakeholders and community partners.

Overall, the implementation of PAThrive is expected to transform the existing manual and fragmented process of training documentation and monitoring into a structured, accessible, and efficient digital system. The project enhances the ability of SLSU to manage extension programs, ensures accurate record-keeping, improves reporting, and strengthens transparency and accountability. PAThrive will also support the continuous improvement of the university's community engagement initiatives and sets the stage for sustainable innovation, collaboration, and evidence-based decision-making in future extension programs.

Background of the Study

The continuous modernization of technology has significantly transformed the way individuals live, learn, and participate in economically related activities. Over the past decades, digital innovations have drastically reshaped the states of professional environments, redefined employment structures, and altered the methods through which individuals acquire skills necessary for different fields in the professional industry. Technological advancements have not only served as the main method of communication and information dissemination but have also influenced how institutions deliver services and implement development-oriented programs within their operations. As industries increasingly adapt to digital systems, the demand for individuals to develop new competencies and continuously upgrade their skills also grows larger in order to remain competitive in dynamic labor markets (Basu & Gupta, 2022).

In the context of employment and professional services, the integration of technology has introduced more efficient methods in terms of recruitment, training, monitoring, and evaluation.

The majority of the digital systems have enabled multiple advancements within institutions such as centralizing their data management, enhancing coordination among different project stakeholders, and reducing inefficiencies associated with their traditional manual operation processes (Fernando et al., 2023). However, despite the advancements, many local institutions and other community-based organizations are still relying on semi-manual or hybrid systems in managing their training services and development programs. These approaches often pose a major factor that result in unorganized handling of documentations, conflicts related to scheduling, reporting delays, and limited accessibility for most stakeholders.

In higher education institutions, Southern Luzon State University (SLSU) specifically, the extension services function as vital mechanisms for bridging academic expertise and community needs. Through its Office of Extension Services under the Office of the Vice President for Research, Extension, Production, Development, and Innovation, implements various community development programs including workshops, seminars, and technical-vocational training. The following initiatives within the institutions are mostly conducted in partnership with local government units (LGUs), non-government organizations, and other sponsoring agencies. The primary objective of these programs is to uplift socio-economic conditions by enhancing participants' technical competencies and preparing them for certification examinations such as NC II assessments.

The Office of the Extension Services mainly targets different groups of beneficiaries including out-of-school youth, displaced workers, solo parents, farmers, retirees, and other community members within the prescribed age bracket on their training programs. The implementation of the programs involves different comprehensive planning processes such as needs assessment surveys, proposal preparations, scheduling reviews, board approvals, memorandum of agreement (MOA) facilitation, fund allocation, and impact assessment. The majority of the documentation and data handling activities remain paper-based or managed through limited digital platforms. Specific activities like manual screening, generation of reports, scheduling within parties, and assessment handling contains multiple operational challenges when managing multiple programs conducted across the different operational quarters in particular.

Shukla (2023) emphasizes that centralized digital systems play a crucial role in improving collaborations with different institutions, minimizing redundancy within the processing of information, and enhancing reporting accuracy. The application of structured information systems enables real-time tracking of participants, trainers, program schedules, and evaluation results, leading to a more efficient workflow within the organization's regular operations. In extension and community engagement settings, digital platforms can also support promotional activities, information dissemination, and communication within different stakeholders, ensuring that beneficiaries and partner organizations remain informed of announcements and other training opportunities.

Objectives of the Study

The main objective of the project is to develop a centralized and efficient web-based system for the Office of the Extension Services of Southern Luzon State University that supports the management of extension training implementation and post-training monitoring. The system aims to improve documentation, tracking, and reporting of extension activities through a structured and role-based digital platform.

Specifically, the researchers sought to:

1) Design and develop PAThrive: A Web-Based Training Management and Skills Utilization Tracking System for Office of the Extension Services of Southern Luzon State University which will be capable of:
    a) providing a centralized platform for recording and managing approved extension trainings during their implementation stage across different academic departments of the university
    b) enabling trainers to digitally encode and upload training modules, presentation materials, and other implementation-related documents
    c) tracking post-training evaluations and skill utilization data within the scope of the Extension Office
    d) publishing announcements and updates related to ongoing and completed trainings, including a public viewing feature for selected accomplishments
    e) generating structured reports such as training accomplishment reports, participant summaries, department-level reports, and consolidated impact summaries.
2) Evaluate the PAThrive system in accordance with ISO/IEC 25010:2023 software quality standards in terms of:
    a) Functional Suitability
    b) Performance Efficiency
    c) Compatibility
    d) Interaction Capability
    e) Reliability
    f) Security

g) Maintainability

h) Safety

i) Flexibility

3) Implement and deploy the PAThrive system within the Office of the Extension Services of Southern Luzon State University, ensuring that the system is fully operational and accessible to users.

Significance of the Study

The proposed system is expected to benefit several stakeholders involved in community extension and training programs such as:

Office of Extension Services. TThe system will provide the Office of Extension Services with a centralized and organized digital repository for training-related records, including participant profiles, activity schedules, documentation, assessment results, and implementation reports by reducing dependence on paper-based and scattered digital files, the platform may help minimize manual document consolidation and improve the accuracy and timeliness of report preparation. The structured data stored in the system may also support monitoring and documentation of training implementation outcomes and facilitate compliance with institutional and partner reporting requirements.

College Extension Coordinators and Trainers. Extension coordinators and trainers may benefit from having a structured system for encoding and uploading training-related documents during and after the conduct of programs. Trainers can systematically upload modules, presentation materials, and evaluation results, while coordinators can submit impact assessment summaries and monitor departmental records. The system may help simplify record keeping, reduce duplication of files, and ensure that required documents are properly archived.

Partner Local Government Units (LGUs) and Affiliated Organizations. Partner institutions collaborating with the university in extension programs may benefit from clearer and more organized documentation of training outputs and summarized impact results. Although external partners do not have direct access to internal system functions, they may view selected accomplishments and completed trainings through the public viewing feature. The Contact Us feature also allows interested agencies to directly communicate with the Extension Office. These features may help strengthen transparency, improve institutional visibility, and encourage future partnerships and collaborative training initiatives.

Training Participants and Beneficiaries. Community members who participate in livelihood and skills development programs may indirectly benefit from the system through the proper documentation and maintenance of training records by program coordinators. The platform ensures that participant training histories, completed activities, and assessment results are accurately stored for institutional reference. Although participants do not directly access the system, reliable documentation may support verification processes for future employment opportunities, certification requirements such as NC II examinations, or other livelihood-related references. Additionally, the system may improve the dissemination of training announcements and program information, increasing awareness and accessibility of extension activities.

University Administration. The university administration may use system-generated reports and performance summaries as reference materials in evaluating extension training implementation outcomes. The platform supports monitoring of participation data, training outputs, and assessment summaries without directly providing decision-making guidance on institutional policies. It also contributes to the university's goal of improving extension service management by promoting organized and technology-enabled documentation of community development activities.

Future Researchers and System Developers. This study may serve as a reference for future research and development projects focusing on digital transformation in extension services, training management, and community engagement systems. The findings and system framework may provide insights into best practices in organizing, monitoring, and evaluating institutional training programs within higher education settings or similar organizations.

Scope and Limitation

The PAThrive system is a web-based application developed to support the implementation and post-training monitoring of extension programs under the Office of the Extension Services of Southern Luzon State University. It is designed to make training documentation and monitoring more organized by replacing paper-based processes with a centralized and role-based digital system. Through the proper recording of training activities, document submissions, evaluations, and impact assessment results, the system helps ensure that extension programs are properly documented, monitored, and reported.

The system has three user levels, namely the Super Admin, Admin, and End User. Each user has specific responsibilities based on their role. When users log in, they are directed to a dashboard where they can view relevant training information. The Super Admin, usually the Director of the Extension Services, has full access to the system. This role oversees all departmental training records, monitors consolidated accomplishment and impact summaries, manages user accounts, and generates overall reports for the Extension Office.

The Admin, who serves as the extension coordinator of each department, manages training records within their assigned department. The Admin encodes approved training details such as the title, schedule, venue, implementing department, and participants. After the training and necessary follow-up activities are conducted, the Admin submits the impact assessment summaries based on collected data. The Admin also monitors document submissions and generates department-level reports when needed.

The End Users, composed of trainers, are responsible for encoding and uploading training implementation data. They upload the training modules, presentation materials, and other instructional documents used during the conduct of the training. Trainers also encode post-training evaluations and other required implementation documents. All submitted records are stored digitally in the system and can be retrieved for monitoring and reporting purposes.

The system also provides an announcement and public viewing feature. Authorized users may post updates regarding ongoing or completed trainings. External users can view selected information such as completed trainings and accomplishment summaries. They may also send inquiries through the Contact Us feature, which redirects messages to the system. However, external users are not allowed to edit records or access internal monitoring and reporting tools.

Despite its features, the PAThrive system has certain limitations. It only covers activities during and after the conduct of training. It does not include training planning, proposal preparation, budgeting, approval workflows, or scheduling authorization. The system depends on manual data entry, which means the accuracy of reports and impact assessments relies on the completeness and correctness of the information entered by authorized users. Since it is fully web-based, it requires a stable internet connection and access to a compatible web browser. It does not function offline and does not integrate with other university systems such as enrollment, payroll, research management, or human resource systems. The system is intended solely for extension training implementation and post-training monitoring within the university.

Definition of Terms

Administrator (Admin). Refers to the designated Extension Coordinator of a specific department in the PAThrive system who is authorized to encode approved training details, monitor document submissions, submit impact assessment summaries, and generate department-level reports within the system.

Beneficiaries. Refers to the identified participants of extension trainings conducted by the Office of Extension Services of Southern Luzon State University (SLSU), including but not limited to out-of-school youth (OSY), solo parents, displaced workers, farmers, retirees, and other qualified community members who receive technical and livelihood skills training.

Centralized Repository. Refers to the unified digital storage within PAThrive where all training implementation records, instructional materials, post-training evaluations, and impact assessment summaries are systematically stored and retrieved for monitoring and reporting purposes.

Compatibility. Refers to the degree to which PAThrive can operate effectively across different web browsers and devices within the required operating environment, as evaluated under ISO/IEC 25010:2023 standards.

Contact Us Feature. Refers to the built-in public inquiry tool of PAThrive that allows external stakeholders to send messages directly to the Office of Extension Services for training-related concerns and partnership opportunities.

End User. Refers to authorized trainers in PAThrive who are responsible for uploading training modules, presentation materials, and encoding post-training evaluation data and other implementation-related documents.

Extension Office. Refers to the official administrative office under SLSU responsible for planning, coordinating, implementing, monitoring, and documenting university-wide extension and community engagement programs. In this study, it serves as the primary implementing unit and end-user environment of the PAThrive system.

Extension Services. Refers to the university-wide community engagement programs of SLSU that deliver training, workshops, and technical-vocational initiatives in partnership with Local Government Units (LGUs) and other stakeholders.

Functional Suitability. Refers to the extent to which PAThrive provides functions that meet the specified requirements for managing training implementation and post-training monitoring, based on ISO/IEC 25010:2023 evaluation criteria.

Impact Assessment. Refers to the structured process conducted after the completion of a training program to determine the extent to which beneficiaries applied the skills acquired, including employment status, certification outcomes (e.g., NC II), and livelihood engagement. In this study, impact assessment data are encoded and summarized within the PAThrive system.

Impact Assessment Summary. Refers to the consolidated digital report generated within PAThrive that presents analyzed results of post-training monitoring and skill utilization of beneficiaries.

Interaction Capability. Refers to the usability and user interface quality of PAThrive that enables Super Admins, Admins, End Users, and external viewers to interact efficiently and accomplish assigned tasks within the system.

ISO/IEC 25010:2023. Refers to the international software quality standard used in this study as the basis for evaluating PAThrive in terms of functional suitability, performance efficiency, compatibility, interaction capability, reliability, security, maintainability, safety, and flexibility.

Local Government Unit (LGU). Refers to government agencies at the municipal, city, or provincial level that partner with SLSU in implementing extension training programs and referring qualified beneficiaries.

Module. Refers to the structured instructional material prepared and used by trainers during the conduct of extension training programs, which may include lesson guides, presentation slides, handouts, and activity materials uploaded and stored in PAThrive for documentation purposes.

PAThrive. Refers to the proposed web-based Training Management and Skills Utilization Tracking System developed in this study to centralize documentation, monitoring, reporting, and public viewing of extension training implementation under the Office of the Extension Services of SLSU.

Paper-Based System. Refers to the traditional method of managing training documents using printed forms, hard-copy modules, and manually consolidated reports, which may result in delays, fragmented records, and increased administrative workload.

Performance Efficiency. Refers to the capability of PAThrive to provide appropriate response times, processing speed, and resource utilization when managing training records and generating reports, based on ISO/IEC 25010:2023.

Public Viewing Feature. Refers to the limited-access component of PAThrive that allows external users to view selected completed trainings, accomplishment summaries, and official announcements without accessing internal system data.

Reliability. Refers to the ability of PAThrive to consistently perform its intended functions without system failure under specified conditions for a defined period of time.

Safety. Refers to the degree to which PAThrive minimizes potential risks to data integrity and prevents unintended negative impacts on stored training records during system use.

Security. Refers to the protection mechanisms embedded in PAThrive that safeguard user accounts, training records, and confidential data from unauthorized access or modification.

Skills Utilization Tracking. Refers to the system feature in PAThrive that records and monitors how beneficiaries apply the skills acquired from extension trainings for employment, certification, or livelihood activities.

Super Administrator (Super Admin). Refers to the highest-level authorized user in PAThrive, typically the Director of the Office of the Extension Services, who has full access to all departmental records, user account management, consolidated reports, and overall system monitoring functions.

Training Implementation. Refers to the actual conduct of approved extension training programs, including delivery of modules, participation of beneficiaries, documentation of activities, and encoding of post-training evaluation data within PAThrive.

Training Management. Refers to the structured digital process within PAThrive for recording, organizing, monitoring, and reporting training-related information during and after program implementation.

Web-Based System. Refers to an application that operates through an internet browser and requires a stable internet connection for access, data entry, document uploads, and report generation, as designed in the PAThrive system.